

ARTIFICIAL INTELLIGENCE → MACHINE LEARNING → DATA SCIENCE

The three-course progression at a glance • BSIT, Fall 2026 • The Islamia University of Bahawalpur, Department of Information Technology



WHO OWNS EACH SHARED TOPIC OWNS taught here, with labs and assessment • signpost named and pointed at, deliberately not taught • recap assumed known and revised, not taught from zero

Course	Search & Agents	Neural Networks	Natural Language	RL / MDP	Generative & Agentic	Bayesian Reasoning	Ethics & Responsibility
AI Semester 3	OWNS Ch 2, 4–9	signpost Ch 17: the perceptron and the XOR limit	OWNS Ch 18, the symbolic half: grammars, parsing, Eliza	signpost Ch 17: one agent–reward figure	signpost Ch 20: named as a trend	OWNS Ch 15, inference in a given network	OWNS Ch 21 can its reasoning be trusted and audited?
ML Semester 6	—	OWNS Ch 13, perceptron to backpropagation	—	OWNS Ch 18–20, HMMs, MDPs, Q-learning	—	OWNS Ch 10, learning the parameters	OWNS Ch 21 is the learned model fair?
DS Semester 7	—	recap Ch 17–18, architectures at scale	OWNS Ch 20, the neural half: embeddings, transformers	—	OWNS Ch 21–22, generative and agentic systems	OWNS Ch 6, description: distributions and priors	OWNS Ch 28 privacy, security, duty to the subject

The distinction is not the technique — it is where the rules come from. Artificial Intelligence is told the rules and must work out what to do, so it can hand you a plan, a proof or an inference chain; it is deliberately symbolic and stops at the perceptron. Machine Learning is given examples and must work out the rule, so it hands you a model and an error estimate. Data Science is given a question and messy data and must work out what is true, so it hands you a decision and a caveat. Why AI teaches no machine learning: Chapter 17 covers the three learning paradigms the HEC 2017 outline names, works the perceptron numerically, shows why one perceptron cannot represent exclusive-or, and hands over — a signpost, not a unit. Appendix F of the AI handout states each boundary chapter by chapter, including the fourth, with Knowledge Representation & Reasoning (HEC 2023, 3 (3-0), prerequisite Artificial Intelligence), which takes description logic, ontology languages, commonsense reasoning and explainable AI further than AI Chapters 10–16 do. Part III of Data Science is dashed because it belongs to Machine Learning: those six chapters are recapped rather than taught from zero, and only feature engineering and anomaly detection are genuinely new. Curriculum basis: the HEC 2017 and HEC 2023 published outlines and the HEC 2025 course learning outcomes, as collected in HEC AI-DS Curriculum Map/.